

Generative AI Analysis and Ethics Report

Project: Unconditional generation of 8x8 handwritten-digit images with a variational autoencoder (VAE).

Overview. The system learns a probability distribution over small grayscale handwritten digits and samples novel digit-like images from it. A VAE is appropriate because it jointly learns a latent representation and a smooth generative decoder; a unit-Gaussian prior enables direct sampling (Kingma & Welling, 2014). This is a learning exercise, not a system for identification, authentication, or handwriting verification.

The accompanying `generative_model.ipynb` contains the complete implementation, data inspection, architecture, training log, loss diagnostic, generated samples, and held-out reconstructions. All reported numerical observations below come from that executed notebook.

Dataset or Prompt Description

The project uses the UCI Optical Recognition of Handwritten Digits dataset, created from handwritten digits written by 43 people and preprocessed into 8x8 integer intensity grids (Alpaydin & Kaynak, 1998). The included files contain 3,823 training examples and 1,797 independent test examples. Each row consists of 64 pixel values from 0 to 16 followed by a digit class label.

The notebook normalizes pixel values to the 0 to 1 range. It uses labels solely to show that every digit category is represented; labels are not input to the VAE. This makes the model unconditional: it can sample a digit-like image but cannot be asked for a specific digit. DATASET.md records the source and download date, and the raw data files are included so notebook execution does not require a network download.

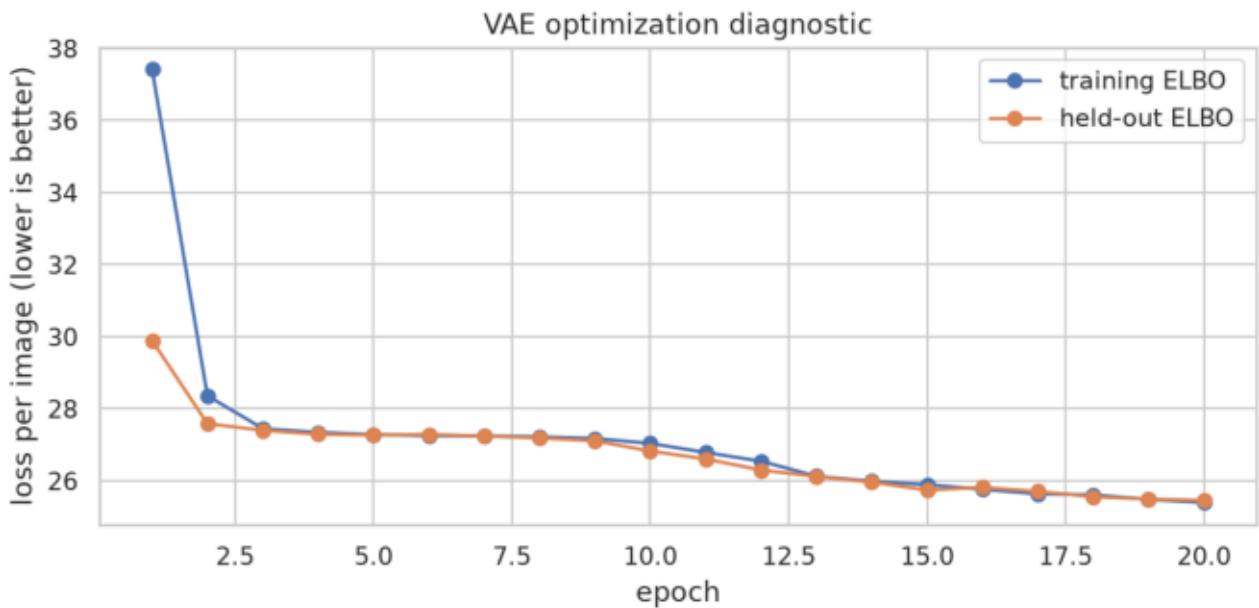
The dataset is suitable for an academic baseline because it is real, public, small enough for local training, and has independent held-out examples. Its low resolution also bounds what visual quality can reasonably be expected.

Model Design and Training Approach

The encoder accepts 64 pixels and uses fully connected layers of 128 and 64 ReLU units. It produces the mean and log variance of a 12-dimensional latent Gaussian. Sampling uses the reparameterization trick. The decoder mirrors the encoder (12 to 64 to 128 to 64) and ends with a sigmoid activation.

Training minimizes summed binary-cross-entropy reconstruction loss plus the analytic Kullback-Leibler divergence between the encoded distribution and a unit Gaussian, the standard VAE objective (Kingma & Welling, 2014). Adam used a learning rate of 0.001, batch size 128, a fixed random seed of 42, and 20 epochs. The test split was evaluated but never used to update parameters.

Observed training loss per image fell from 37.42 in epoch 1 to 25.39 in epoch 20. Held-out loss ended at 25.45; the close final train/test values are a useful diagnostic, though not a direct perceptual-quality measure.



Output Evaluation and Interpretation

The notebook samples 24 latent vectors independently from the unit-Gaussian prior and decodes them. The outputs show recurring loops, vertical strokes, and diagonal/curved marks that resemble components of handwritten digits. The reconstruction comparison also shows that the model often retains central structure and stroke direction for held-out inputs.

A concrete failure case is visible in the generated grid: some samples have low contrast, disconnected strokes, or combine a loop with an incomplete vertical or curved stroke. Such images can be ambiguous between several digits rather than clearly readable. This behavior is consistent with averaging pressure from the reconstruction loss, the small 8x8 images, and unconditional sampling. Therefore this baseline demonstrates plausible digit-like generation, not reliable class-specific synthesis.

Evaluation combines the loss trajectory with qualitative inspection. This is appropriate here because generative quality is not fully captured by one scalar objective; the evaluation claims are deliberately limited to the notebook images rather than claiming classification accuracy or human realism.

24 novel VAE samples from the prior



Ethical Considerations and Responsible Use

The data contain handwriting from only 43 contributors and provide no demographic metadata. The model may therefore encode an unmeasured and narrow subset of handwriting styles. Its digit-like outputs should not be interpreted as representative of populations, accessibility needs, languages, or real-world writing conditions. Documentation of dataset composition and intended use is important for surfacing such gaps (Gebru et al., 2021).

Although images are only 8x8 digits and not signatures, a generative handwriting system could be misrepresented as authentic writing or used to make synthetic examples look human-authored. This project avoids identity claims, does not train on signatures or personal text, and should be presented as educational synthetic imagery. Any downstream use in authentication, evidence, grading, or fraud detection would require substantially different data, validation, security review, consent considerations, and explicit safeguards.

Creative ownership risk is comparatively low for these highly abstract digits, but provenance remains important: the report names the dataset and does not claim generated samples are a particular contributor's handwriting.

Limitations, Future Improvements, and References

Limitations and future improvements. The resolution is only 8x8, training is short, and the fully connected VAE lacks convolutional structure. More data, convolutional encoder/decoder blocks, latent-dimension and KL-weight tuning, and repeated seeds could improve sharpness and stability. A conditional VAE could make class requests possible, while evaluation could add blinded human readability assessment and a classifier-based diversity/coverage analysis. These extensions must still evaluate representational bias and misuse risk.

References

Alpaydin, E., & Kaynak, C. (1998). Optical Recognition of Handwritten Digits [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C50P49>

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12), 86–92. <https://doi.org/10.1145/3458723>

Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. *International Conference on Learning Representations*. <https://arxiv.org/abs/1312.6114>